

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

pd.set_option("display.max_columns", None)

df = pd.read_excel("/content/Online Retail.xlsx")

print("Rows:", df.shape[0])
print("Columns:", df.shape[1])

df.head()

```

Rows: 541909
Columns: 8

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice
0	536365	85123A	WHITE HANGING HEART T- LIGHT HOLDER	6	2010-12-01 08:26:00	2.55
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39

```
df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 541909 entries, 0 to 541908
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype  
---  -
0   InvoiceNo        541909 non-null object  
1   StockCode       541909 non-null object  
2   Description      540455 non-null object  
3   Quantity        541909 non-null int64   
4   InvoiceDate      541909 non-null datetime64[ns]
5   UnitPrice       541909 non-null float64  
6   CustomerID      406829 non-null float64  
7   Country         541909 non-null object  
dtypes: datetime64[ns](1), float64(2), int64(1), object(4)
memory usage: 33.1+ MB

```

```
df.describe(include="all")
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate
count	541909.0	541909	540455	541909.000000	541909
unique	25900.0	4070	4223	NaN	NaN
top	573585.0	85123A	WHITE HANGING HEART T- LIGHT HOLDER	NaN	NaN
freq	1114.0	2313	2369	NaN	NaN
mean	NaN	NaN	NaN	9.552250	2011-07-04 13:34:57.156386048
min	NaN	NaN	NaN	-80995.000000	2010-12-01 08:26:00
25%	NaN	NaN	NaN	1.000000	2011-03-28 11:34:00
50%	NaN	NaN	NaN	3.000000	2011-07-19 17:17:00
75%	NaN	NaN	NaN	10.000000	2011-10-19 11:27:00
max	NaN	NaN	NaN	80995.000000	2011-12-09 12:50:00
std	NaN	NaN	NaN	218.081158	NaN

```
df.isnull().sum()
```

	0
InvoiceNo	0
StockCode	0
Description	1454
Quantity	0
InvoiceDate	0
UnitPrice	0
CustomerID	135080
Country	0

dtype: int64

```
df.duplicated().sum()
```

```
np.int64(5268)
```

```
print("Duplicate rows:", df.duplicated().sum())

print("\nMissing values:")
print(df.isnull().sum())

print("\nNegative quantities:")
print((df["Quantity"] < 0).sum())

print("\nZero/negative prices:")
print((df["UnitPrice"] <= 0).sum())

print("\nCancelled invoices:")
print(df["InvoiceNo"].astype(str).str.startswith("C").sum())
```

Duplicate rows: 5268

Missing values:

InvoiceNo	0
StockCode	0
Description	1454
Quantity	0
InvoiceDate	0
UnitPrice	0
CustomerID	135080
Country	0

dtype: int64

Negative quantities:
10624

Zero/negative prices:
2517

Cancelled invoices:
9288

```
df_clean = df.copy()
```

```
df_clean = df_clean.drop_duplicates()

print("Rows after removing duplicates:", len(df_clean))
```

Rows after removing duplicates: 536641

```
df_clean = df_clean.dropna(subset=["Description"])

print("Rows after removing missing descriptions:", len(df_clean))
```

Rows after removing missing descriptions: 535187

```
df_clean["InvoiceNo"] = df_clean["InvoiceNo"].astype(str)

df_clean = df_clean[
    ~df_clean["InvoiceNo"].str.startswith("C")
]

print("Rows after removing cancellations:", len(df_clean))
```

```
Rows after removing cancellations: 525936
```

```
df_clean = df_clean[df_clean["Quantity"] > 0]
```

```
df_clean = df_clean[df_clean["UnitPrice"] > 0]
```

```
df_clean["Revenue"] = (  
    df_clean["Quantity"] * df_clean["UnitPrice"]  
)
```

```
df_clean[[  
    "InvoiceNo",  
    "Description",  
    "Quantity",  
    "UnitPrice",  
    "Revenue"  
]].head()
```

	InvoiceNo	Description	Quantity	UnitPrice	Revenue
0	536365	WHITE HANGING HEART T-LIGHT HOLDER	6	2.55	15.30
1	536365	WHITE METAL LANTERN	6	3.39	20.34
2	536365	CREAM CUPID HEARTS COAT HANGER	8	2.75	22.00
3	536365	KNITTED UNION FLAG HOT WATER BOTTLE	6	3.39	20.34
4	536365	RED WOOLLY HOTTIE WHITE HEART.	6	3.39	20.34

```
df_clean["InvoiceDate"] = pd.to_datetime(  
    df_clean["InvoiceDate"]  
)
```

```
df_clean["Year"] = df_clean["InvoiceDate"].dt.year  
df_clean["Month"] = df_clean["InvoiceDate"].dt.month  
df_clean["MonthName"] = df_clean["InvoiceDate"].dt.month_name()  
df_clean["YearMonth"] = df_clean["InvoiceDate"].dt.to_period("M").astype(str)  
df_clean["DayOfWeek"] = df_clean["InvoiceDate"].dt.day_name()  
df_clean["Hour"] = df_clean["InvoiceDate"].dt.hour
```

```
print("Original rows:", len(df))
print("Clean rows:", len(df_clean))
print("Rows removed:", len(df) - len(df_clean))

print("\nCleaned shape:")
print(df_clean.shape)

print("\nMissing values:")
print(df_clean.isnull().sum())

print("\nRevenue statistics:")
print(df_clean["Revenue"].describe())
```

```
Original rows: 541909
Clean rows: 524878
Rows removed: 17031
```

```
Cleaned shape:
(524878, 15)
```

```
Missing values:
```

InvoiceNo	0
StockCode	0
Description	0
Quantity	0
InvoiceDate	0
UnitPrice	0
CustomerID	132186
Country	0
Revenue	0
Year	0
Month	0
MonthName	0
YearMonth	0
DayOfWeek	0
Hour	0

```
dtype: int64
```

```
Revenue statistics:
```

count	524878.000000
mean	20.275399
std	271.693566
min	0.001000
25%	3.900000
50%	9.920000
75%	17.700000
max	168469.600000

```
Name: Revenue, dtype: float64
```